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基於子問題熵驅動之高效檢索增強生成

SEER: Sub-question Entropy-driven Efficient Retrieval

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Abstract

This study proposes SEER (Sub-question Entropy-driven Efficient Retrieval), a sub-question-level adaptive Retrieval-Augmented Generation (RAG) framework. Existing RAG systems typically apply a one-size-fits-all retrieval strategy: unnecessary retrieval on simple queries adds noise and cost, while a single retrieval pass on complex multi-hop questions often yields incomplete or incorrect answers.

Recent adaptive RAG methods attempt to address this issue but operate at the wrong granularity: question-level routing is too coarse to react mid-reasoning, while token-level probing is too noisy. Sub-question-level approaches such as DeepRAG match the right granularity but rely on costly binary tree search, SFT, and DPO pipelines, sacrificing portability and interpretability.

The core idea of SEER is: after decomposing a complex question into atomic sub-questions, we compute the entropy over the first-entity span of each tentative answer as a measure of model confidence, and use this signal to gate retrieval. SEER consists of three stages: (1) data construction integrating MuSiQue, HotpotQA, and 2WikiMultiHopQA with quality filtering; (2) supervised fine-tuning of a lightweight decomposer; and (3) a training-free entropy-threshold gate. We expect SEER to match DeepRAG's Exact Match on multi-hop QA benchmarks while substantially reducing training cost and improving interpretability.

Keywords: Retrieval-Augmented Generation, Adaptive Retrieval, Multi-hop Question Answering, Entropy, Sub-question Decomposition, Large Language Models

摘要

本研究提出 SEER (Sub-question Entropy-driven Efficient Retrieval)，一個基於子問題層級的自適應檢索增強生成 (Retrieval-Augmented Generation, RAG) 框架。現有 RAG 系統在面對不同複雜度的查詢時，往往採用一體適用的檢索策略：對簡單問題進行不必要的檢索會引入雜訊並增加成本；對複雜的多跳推理問題僅進行單次檢索則經常導致答案不完整或錯誤。

近期的自適應 RAG 方法雖試圖解決此問題，但其決策粒度往往不適切：問題層級的路由過於粗糙，無法在推理過程中即時調整；詞元 (token) 層級的偵測則過於嘈雜。雖有方法採用子問題層級的決策 (如 DeepRAG)，但仰賴二元樹搜尋、SFT 與 DPO 等繁複訓練流程，犧牲了可攜性與可解釋性。

本研究的核心想法是：將複雜問題分解爲原子化子問題後，針對每個子問題答案中的「首實體跨度 (first-entity span)」計算熵值，作爲模型自信度的指標，再以此決定是否觸發檢索。SEER 包含三階段：(1) 資料建構，整合 MuSiQue、HotpotQA 與 2WikiMultiHopQA 資料集並進行品質過濾；(2) 以 SFT 訓練輕量化分解器；(3) 訓練無關的熵閾值門控機制。我們預期 SEER 在多跳問答基準上能達到與 DeepRAG 相當的 Exact Match 表現，同時顯著降低訓練成本並提升可解釋性。

關鍵字：檢索增強生成、自適應檢索、多跳問答、熵、子問題分解、大型語言模型

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Chapter 1 Introduction

1.1 Background and Motivation

Large language models (LLMs) often hallucinate when queried about recent, obscure, or proprietary information outside their training scope. Retrieval-Augmented Generation (RAG) addresses this by fetching relevant external data, keeping the model updated and significantly reducing factually incorrect outputs [1], [2]. However, applying RAG indiscriminately to every question creates two main problems: for simple queries, unnecessary retrieval adds noise and increases latency; for complex multi-hop queries, a single retrieval step is often insufficient and yields incomplete answers.

1.2 Research Objectives

This thesis introduces SEER (Sub-question Entropy-driven Efficient Retrieval), a sub-question-level adaptive retrieval framework that:

- Decomposes a complex question into atomic sub-questions via a lightweight SFT decomposer.
- Probes the model's confidence on each sub-question by computing entropy over the first-entity span of its tentative answer.
- Gates retrieval per sub-question — low entropy uses parametric memory, high entropy invokes the retriever.

Chapter 2 Related Work

2.1 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) [1] augments a parametric language model with a non-parametric retriever, fetching external passages at inference time and concatenating them with the input prompt. The paradigm mitigates hallucination on long-tail and time-sensitive facts and underpins production assistants such as Bing Chat and Perplexity. However, the default "always retrieve" policy has two well-documented failure modes:

- **Noise injection on easy queries.** For questions the model already knows (e.g. "Who wrote *Hamlet*?"), retrieved passages introduce distractor entities that can override correct parametric knowledge [2].
- **Insufficient evidence on hard queries.** For multi-hop questions, a single retrieval pass rarely surfaces all bridging entities, leading to incomplete or compositionally wrong answers [3].

These two failure modes motivate the three themes below: *adaptive* retrieval addresses *when* to retrieve, *reasoning*-based RAG addresses *how* to retrieve, and the *knowledge boundary* literature studies the signal that a good gating decision must ultimately rely on.

2.2 Adaptive Retrieval-Augmented Generation

Adaptive RAG methods decide per query (or per reasoning step) whether and how much to retrieve. They can be characterised along three axes: the **decision granularity** (question / sentence / token / sub-question), the **signal** used to gate retrieval, and the **training cost** (training-free / SFT / RL or DPO). Organising by the gating signal — the most consequential axis for SEER — yields three families: *classifier-based*, *confidence-based*, and *reasoning- and policy-based* gating.

2.2.1 Classifier-Based Gating

The first family trains a dedicated discriminator that inspects the query and emits a retrieval decision.

Adaptive-RAG [4] routes each incoming query to one of three strategies based on its complexity: (i) *no retrieval* — the LLM answers from parametric memory only; (ii) *single-step retrieval* — one BM25 or dense retrieval pass followed by reading; and (iii) *multi-step retrieval* — iterative retrieve-and-reason (e.g. IRCoT [3]). The routing decision is produced by a small T5-Large classifier trained on *automatically derived labels*: each training question is labelled with the cheapest strategy whose prediction matches the gold answer, with ties broken by the dataset prior (single-hop datasets bias toward "no retrieval"; multi-hop datasets toward "multi-step"). Evaluated on Natural Questions, TriviaQA, SQuAD, MuSiQue, HotpotQA, and 2WikiMultihopQA, Adaptive-RAG matches a uniform multi-step baseline while approximately halving the average retrieval count. Its chief limitation is that the gate fires *once per query*, before any reasoning has occurred: the classifier cannot react to mid-reasoning uncertainty, and the three labels are coarse — a five-hop and a two-hop question both receive "multi-step".

SKR [5] learns a binary "does the model know it?" classifier from the model's own correctness on a calibration set. At inference, a query is classified as "known" (skip retrieval) or "unknown" (retrieve), making SKR effectively a binary version of Adaptive-RAG: cheaper than reflection-token methods, but unable to distinguish single-step from multi-step needs. Both classifier-based methods require an additional trained head and, more fundamentally, commit to a decision before the model has begun to reason.

2.2.2 Confidence- and Uncertainty-Based Gating

The second family replaces the external classifier with an *intrinsic* uncertainty signal read off the generation process, and is typically training-free.

FLARE [6] pushes the gating granularity from question level to sentence level. At each generation step it first drafts a tentative next sentence; if any token in that sentence carries a probability below a threshold θ, the low-probability span is masked, the sentence

is re-issued as a retrieval query, and generation is restarted with the retrieved evidence. The method is training-free and language-agnostic, but token probability is a noisy proxy for factual uncertainty (common-word low-probability spans trigger spurious retrievals), and sentence-level rewriting interacts poorly with chain-of-thought, whose natural reasoning unit is a sub-question rather than a sentence.

DRAGIN [7] extends FLARE with *Real-time Information Need Detection* (RIND) and *Query Formulation based on Self-attention* (QFS). RIND scores each generated token by combining token entropy, attention weight to subsequent tokens, and semantic content, triggering retrieval when the score exceeds a threshold; QFS then builds the query from the most-attended tokens in the preceding context rather than the raw lookahead. DRAGIN is training-free and improves over FLARE on 2WikiMultihopQA, HotpotQA, and IIRC, but inherits FLARE's token-level granularity mismatch and adds attention-extraction overhead.

SeaKR [8] taps the residual stream directly, estimating uncertainty by aggregating the variance of hidden states across decoding steps; high variance triggers retrieval. The authors argue that internal variance captures factual uncertainty more faithfully than output-layer entropy, and add a re-ranking step that prefers passages which reduce hidden-state variance. SeaKR is reported as state-of-the-art among training-free methods on Natural Questions, HotpotQA, and 2WikiMultihopQA. Across this family, however, the gate ultimately depends on a hand-set threshold over a noisy uncertainty proxy — the central tension SEER seeks to resolve with a sub-question-level entropy probe.

2.2.3 Reasoning- and Policy-Based Gating

The third family folds the retrieval decision into the model's own generation, learning it as part of a reasoning policy.

Self-RAG [9] trains the LLM to emit special *reflection tokens* — `[Retrieve]` (gate), `[IsRel]` (passage relevance), `[IsSup]` (span support), and `[IsUse]` (utility) — by fine-tuning on GPT-4-distilled traces, then performs token-conditioned tree decoding at inference. It outperforms retrieve-then-read baselines on TriviaQA, PopQA, and

ALCE long-form generation, but requires a costly distillation corpus and a custom decoder that interleaves reflection-token sampling with content generation.

DeepRAG [10] models retrieval-augmented reasoning as a Markov Decision Process: at each sub-query the agent chooses between answering from parametric memory and issuing a retrieval call. Training has two stages — *Stage I* uses binary tree search to roll out both branches per sub-query, producing a balanced SFT corpus that teaches both "skip" and "retrieve" patterns; *Stage II* (*Chain of Calibration*) applies DPO-style preference learning over the traces. Trained on 4K HotpotQA + 1K 2WikiMultihopQA examples (≈10K via tree augmentation) and evaluated out-of-distribution on PopQA, CAG, WebQuestions, and MuSiQue, DeepRAG achieves the strongest sub-question-level gating to date — but at substantial cost: tree search is $O(2^H)$ in the number of hops, the DPO stage requires careful pair construction, and the gate is a learned implicit policy with no externally interpretable score.

Orthogonal to the gating signal, a complementary line of work treats the retrieval *count* as a user-controllable knob: **Fast-or-Better** [11] exposes a Pareto frontier between latency and accuracy rather than a single operating point, underscoring that adaptive retrieval is fundamentally a cost–accuracy trade-off. SEER positions itself within this family in spirit — sub-question-level gating like DeepRAG — but replaces the learned policy with an interpretable, training-cheap entropy probe, recovering the transparency of the confidence-based methods without their token-level granularity mismatch.

2.3 Reasoning in Retrieval-Augmented Generation

A parallel thread focuses on *how* to retrieve and reason once retrieval is deemed necessary, independent of the gate.

The foundational technique is **IRCoT** [3], which interleaves chain-of-thought reasoning with retrieval: after each generated CoT sentence, the partial reasoning becomes the next query. IRCoT is the canonical "multi-step" implementation referenced by Adaptive-RAG and DeepRAG, but its retrieval count grows linearly in the CoT length, regardless of whether intermediate steps are factually difficult. **Question decomposition** [12] instead

explicitly splits a multi-hop question into atomic sub-questions before answering, treating decomposition as a separate seq-to-seq problem. SEER inherits this decomposition paradigm but couples it with per-sub-question entropy probing, avoiding both the linear cost of IRCoT and the policy-learning cost of DeepRAG.

More recent work synthesises reasoning data to internalise the retrieve-and-reason loop. **Auto-RAG** [13] autonomously synthesises multi-turn retrieval–reasoning trajectories and fine-tunes on them, so that the model plans its own retrieval steps without hand-crafted prompts. **Search-o1** [14] embeds retrieval inside a large reasoning model in an agentic loop, augmenting o1-style long-chain reasoning with on-demand search and a document-reasoning module that distils retrieved passages before they re-enter the chain. **AirRAG** [15] activates the model's intrinsic reasoning by combining Monte Carlo Tree Search with self-consistency to explore alternative solution paths. These methods substantially strengthen the reasoning side of RAG, but — like DeepRAG — they rely on heavy data synthesis or search-time expansion; SEER instead keeps the reasoning component a single lightweight SFT decomposer and concentrates its novelty in the gating signal.

2.4 Knowledge Boundary

Every adaptive method above ultimately rests on a model's ability to recognise its own *knowledge boundary* — the line between what it reliably knows and what it must look up. A dedicated line of work studies this capacity directly, and we group it into three strands: *self-knowledge elicitation*, *uncertainty estimation*, and the *interaction between the knowledge boundary and retrieval*.

Self-knowledge and its calibration. A first strand asks whether a model can introspect on what it knows. Kadavath et al. [16] show that LLMs are, to a useful degree, calibrated: a model can be prompted to predict the probability `P(True)` that its own answer is correct, and these self-judgements correlate with actual accuracy — though calibration degrades on out-of-distribution and adversarial questions. Yin et al. [17] construct a "SelfAware" benchmark of answerable and unanswerable questions and find that LLMs

only partially "know what they don't know": self-knowledge can be elicited but lags well behind human performance. From the supervised side, Kapoor et al. [18] argue that models "must be taught to know what they don't know", demonstrating that lightweight calibration-tuning or probing substantially sharpens a model's estimate of its own uncertainty. Yin et al. [19] formalise the notion of a knowledge boundary itself, proposing to benchmark a model by probing the frontier of its reliable knowledge rather than by aggregate accuracy alone.

Uncertainty estimation as a boundary signal. A second strand asks how to *measure* the boundary at inference time. A naive token-probability or token-entropy signal conflates two kinds of uncertainty: *epistemic* (the model genuinely does not know) and *lexical* (many surface forms express the same fact). Kuhn et al. [20] introduce *semantic entropy*, which clusters multiple sampled answers by meaning before computing entropy, so that paraphrases of one correct answer do not register as uncertainty; Farquhar et al. [21] scale this idea and show in *Nature* that semantic entropy reliably detects confabulations across QA and biography generation. This line of work is the closest in spirit to SEER's gating signal: both read out the knowledge boundary from generation-time entropy. SEER deliberately adopts the cheaper *token-level* entropy — computed once per sub-question rather than over a cluster of samples — trading a small amount of the semantic-entropy robustness for a single forward pass, which is what makes per-sub-question gating affordable in a multi-hop loop.

The boundary under retrieval. A third strand studies the boundary specifically in the RAG setting. Ren et al. [22] show that retrieval augmentation not only injects evidence but also *reshapes* the boundary: relevant passages let a model answer questions it could not before, while irrelevant or distracting passages can push a previously-correct answer over the boundary into error — precisely the "noise injection on easy queries" failure mode of §2.2. This makes a per-step boundary estimate, rather than a one-shot query-level one, essential when retrieval and reasoning are interleaved.

These three strands form the conceptual foundation of SEER. The failure modes of §2.2 — spurious retrieval on easy queries, missing retrieval on hard ones — are all

symptoms of a poorly localised knowledge boundary. Where confidence-based methods read an uncertainty proxy at the token or sentence level and policy-based methods bury the boundary inside a learned policy, SEER makes the boundary explicit at the *sub-question* level: the token-entropy probe is a transparent scalar estimate of whether the model knows the answer to a given sub-question, directly operationalising the knowledge-boundary view as a tunable, inspectable gate.

2.5 Multi-Hop Question Answering Datasets

The evaluation of adaptive RAG hinges on datasets that span both single-hop and multi-hop complexity. Table 2.1 summarises the four datasets used in this thesis.

2.5.1 HotpotQA

HotpotQA [23] is the canonical multi-hop QA benchmark, containing 113K Wikipedia-based questions requiring 2-hop reasoning. Each example provides (a) the question, (b) the gold answer, (c) two gold-supporting paragraphs out of 10 distractors (in the `distractor` setting), and (d) sentence-level supporting facts. Questions fall into two compositional patterns: *bridge* (the answer to sub-question 1 is the subject of sub-question 2) and *comparison* (two entities are compared along a shared attribute). HotpotQA is a primary training source for DeepRAG and a standard evaluation set for nearly all adaptive RAG papers.

2.5.2 2WikiMultihopQA

2WikiMultihopQA [24] constructs 192K multi-hop questions by leveraging Wikidata knowledge-graph paths, ensuring each question requires traversal of an explicit reasoning chain. It includes four question types (comparison, inference, compositional, and bridge-comparison) and provides KG-triple annotations alongside textual evidence. Compared to HotpotQA, 2WikiMultihopQA exhibits less surface-form leakage between the question and the supporting passages, making it a more reliable test of compositional reasoning.

2.5.3 MuSiQue

MuSiQue (**Mu**ltihop questions via **Si**ngle-hop **Que**stions) [25] addresses two pathologies of earlier datasets: (i) shortcut answers via single-hop reasoning, and (ii) weak distractor selection. MuSiQue is built by composing 2–4 single-hop questions into chains using strict connectivity criteria, yielding 25K questions of 2-hop, 3-hop, and 4-hop varieties. Importantly, MuSiQue provides the *decomposed* sub-questions as supervision, which makes it a natural training source for decomposer models such as the one used in SEER. Its high distractor difficulty makes it the hardest of the four datasets in this thesis.

2.5.4 PopQA

PopQA [26] is a *single-hop* entity-centric QA dataset of 14K questions, each derived from a Wikidata subject–property–object triple. Critically, PopQA tags each entity with its *popularity* (measured by monthly Wikipedia page views), enabling stratified evaluation of factual recall on long-tail entities. Despite being single-hop, PopQA exposes the over-decomposition failure mode of decomposer-based pipelines: trained on multi-hop data, a decomposer may emit two-step plans even when one step suffices, harming both EM and efficiency.

2.5.5 Single-Hop Open-Domain QA

For completeness, prior adaptive-RAG papers also evaluate on **Natural Questions (NQ)** [27], **TriviaQA** [28], and **SQuAD** [29]. These are single-hop datasets whose answers typically exist in a single Wikipedia paragraph, and they primarily test whether an adaptive method can correctly *skip* retrieval rather than orchestrate it. This thesis follows the DeepRAG protocol of evaluating on PopQA as the representative single-hop test set.

2.6 Summary of Related Work

Table 2.2 summarises the key differences between SEER and existing adaptive RAG methods along the three axes introduced above. SEER occupies the under-explored region

Table 2.1: Multi-hop QA datasets used in this thesis. #hops indicates the typical reasoning depth; "Decomp." indicates whether gold sub-questions are provided.

Dataset	Size	#hops	Decomp.	Source
HotpotQA	113K	2	No	Wikipedia (crowd)
2WikiMultihopQA	192K	2	No	Wikipedia + Wikidata
MuSiQue	25K	2–4	Yes	Composed single-hops
PopQA	14K	1	Yes (triple)	Wikidata triples

of *sub-question-level gating with an interpretable, training-cheap signal*: only a single SFT pass is needed for the decomposer (no DPO, no tree search), and the entropy probe is a transparent scalar — a direct estimate of the knowledge boundary (§2.4) — that can be inspected and tuned per dataset.

Table 2.2: Comparison of adaptive RAG methods along granularity, gating signal, interpretability, training cost, and decision unit.

Method	Decomposer	Gating Signal	Interpretable	Training Cost	Gra
Adaptive-RAG [4]	None	Query classifier	Partial	Low (T5-Large)	Qu
FLARE [6]	None	Token probability	Yes	None	Se
Self-RAG [9]	None	Reflection tokens	Partial	High (distill)	S
DRAGIN [7]	None	Entropy + attention	Yes	None	T
SKR [5]	None	Binary self-knowledge	Partial	Low	Qu
SeaKR [8]	None	Hidden-state variance	No	None	Se
DeepRAG [10]	SFT	Learned policy (SFT+DPO)	No	High	S
SEER (Ours)	SFT	Token entropy (τ-gated)	**Yes**	**Low** (SFT only)	**Su**

Chapter 3 Methodology

This chapter presents **SEER** (**S**ub-question **E**ntropy-driven **E**fficient **R**etrieval), a sub-question-level adaptive RAG framework. Section 3.1 gives an end-to-end overview. Sections 3.3–3.6 describe each pipeline stage. Section 3.7 details the LoRA training protocol for the decomposer, and Section 3.8 explains the two inference modes (B / C) used for ablation.

3.1 Overview

Given a question q and a candidate paragraph pool $\mathcal{P}_q$ (the standard multi-hop QA evaluation setup), SEER produces a final answer $\hat{a}$ in four stages:

1. **Decompose.** The decomposer $\mathcal{D}_\theta$ maps q to an ordered list of atomic sub-questions $\{s_1, s_2, \ldots, s_n\}$ in a knowledge-graph-triple notation, with explicit cross-references between steps.

2. **Per-step probe.** For each sub-question s_i, the answerer $\mathcal{A}_\theta$ generates a tentative answer $\tilde{a}_i$ without retrieval and emits a token-level confidence signal σ_i.

3. **Gate.** If $\sigma_i \geq \tau$ (high uncertainty), the BM25 retriever fetches the top-k paragraphs from $\mathcal{P}_q$ for s_i, and $\mathcal{A}_\theta$ re-answers s_i conditioned on the retrieved context to produce a_i. Otherwise $a_i := \tilde{a}_i$ (parametric memory only).

4. **Synthesize.** The synthesizer (same model as $\mathcal{A}_\theta$) takes q together with all sub-answers $\{(s_i, a_i)\}$ and emits the final terse answer $\hat{a}$.

The decomposer and answerer share a single Llama3-8B-Instruct backbone with a LoRA adapter (rank 16) trained on the decomposition task only. All gating happens at inference time — no policy is learned for the retrieve / skip decision.

Fig. 3.1: SEER pipeline: a question is decomposed into KG-triple sub-questions, each probed for token-level entropy, gated against threshold τ, optionally re-answered with retrieved context, and finally synthesized into a terse answer.

3.2 Problem Formulation

Let $\mathcal{M}_\theta$ be a pretrained language model and $r(s, \mathcal{P}) \rightarrow \mathcal{P}_{\text{top-}k}$ a retriever over a paragraph pool $\mathcal{P}$. A retrieval-augmented QA system produces an answer $\hat{a} = \mathcal{M}_\theta(q, c)$ where c is a (possibly empty) context. We define a per-sub-question *retrieval policy* $\pi : \mathcal{S} \rightarrow \{0, 1\}$ which decides, for each sub-question s, whether to invoke the retriever. The optimisation goal is:

$$\min_{\pi} \quad \mathbb{E}_{(q,a^*)}\big[\mathcal{L}(\hat{a}, a^*) + \lambda \cdot \mathbb{E}_{s \in \mathcal{D}(q)}[\pi(s)]\big] ,$$

where $\mathcal{L}$ is an answer-correctness loss (e.g. $1 - \text{EM}$), $\mathcal{D}(q)$ is the decomposition of q, and $\lambda > 0$ trades off accuracy against retrieval cost. SEER does *not* learn π; instead it instantiates π as a simple threshold rule on a model-derived confidence signal, sidestepping the policy-learning cost of DeepRAG [10].

3.3 Stage 1: Sub-Question Decomposer

3.3.1 Output Format

The decomposer is instructed to emit one sub-question per line, prefixed with $\#\langle i \rangle$ `:`. Each sub-question is expressed as a *knowledge-graph triple* fragment:

$$\#i\texttt{:} \quad \langle \textit{entity} \rangle \;\texttt{>>}\; \langle \textit{relation} \rangle$$

When a sub-question depends on the answer of an earlier step, the placeholder $\#j$ (with $j < i$) substitutes for the unresolved entity. For example, the question "*Where was the director of the film X born?*" decomposes to:

```
#1: X >> director
```

`#2: #1 >> place of birth`

At inference, placeholders are resolved sequentially: once a_1 is determined, `#1` in s_2 is rewritten as a_1 before the answerer is invoked. The KG-triple notation has three benefits over free-form natural-language sub-questions: (i) it is compact and consistent across datasets, reducing decoding variance; (ii) it matches the underlying structure of Wikidata-derived datasets (PopQA, 2WikiMultihopQA); and (iii) it makes downstream entropy probing semantically meaningful — the answer span IS the object entity of the triple.

3.3.2 Decomposer Prompt

The decomposer is prompted with a fixed instruction:

> *Decompose the following multi-hop question into an ordered list of atomic sub-questions. Each sub-question must be answerable in a single retrieval/reasoning step. When a sub-question depends on the answer of an earlier one, refer to that step with a `#N` placeholder (e.g. `#1`). Output one sub-question per line, prefixed with `#<index>:`*

3.3.3 Decomposition Examples by Dataset

Different datasets exhibit distinct decomposition topologies (Table 3.1):

3.4 Stage 2: Token-Level Confidence Probe

For each resolved sub-question s_i, the answerer produces a tentative answer $\tilde{a}_i$ by greedy decoding (`do_sample=False`) under a constraint prompt that forces an answer-only emission:

> *Answer the question with the answer phrase only. No explanation, no preamble, no full sentence — just the answer.*

This prompt is critical: by forbidding preamble and explanation, the generated span

Table 3.1: Representative decomposition patterns produced by the SEER decomposer.

Dataset	Question type	Decomposition
PopQA	1-hop fact lookup	`#1: George Rankin >> occupation`
HotpotQA	2-hop bridge	`#1: Film X >> director` `#2: #1 >> nationality`
2WikiMultihopQA	comparison	`#1: A >> date of birth` `#2: B >> date of birth`

MuSiQue (4-hop)	compositional chain	`#1: X >> capital`
		`#2: #1 >> mayor`
		`#3: #2 >> party`
		`#4: #3 >> founder`

IS the answer entity, so token-level confidence probes the model's certainty about the *factual* target rather than about boilerplate. During generation, the per-step softmax distribution $p_t = \mathrm{softmax}(\mathrm{logits}_t)$ is captured for each generated token t, and three confidence signals are computed:

$$\sigma^{\text{max-ent}} = \max_t H(p_t), \qquad \text{(maximum per-token entropy)} \tag{3.1}$$

$$\sigma^{\text{mean-ent}} = \frac{1}{T}\sum_t H(p_t), \qquad \text{(mean per-token entropy)} \tag{3.2}$$

$$\sigma^{\text{min-lp}} = \min_t \log p_t(y_t), \qquad \text{(minimum log-prob of chosen token)} \tag{3.3}$$

where $H(p) = -\sum_v p(v)\log p(v)$ and y_t is the chosen token at step t. EOS tokens are excluded. Higher entropy (or lower log-prob) indicates higher uncertainty. The default signal in SEER is $\sigma^{\text{max-ent}}$, which empirically correlates best with answer-span uncertainty in pilot studies.

3.5 Stage 3: Retrieval Gate

The retrieval decision per sub-question is a thresholded comparison:

$$\pi(s_i) = \mathbb{1}[\sigma_i \geq \tau] \quad \text{for entropy signals (max-ent / mean-ent),}$$

$$\pi(s_i) = \mathbb{1}[\sigma_i \leq \tau] \quad \text{for log-prob signal (min-lp).}$$

When $\pi(s_i) = 1$, the retriever fetches the top-k paragraphs and the answerer re-runs with the retrieved context. When $\pi(s_i) = 0$, the tentative parametric answer $\tilde{a}_i$ is adopted as a_i. The threshold τ is a single, dataset-agnostic hyperparameter — set to $\tau = 0.3$ for max-entropy in all main experiments (justified in Chapter 4). This unified τ design is a deliberate choice: tuning τ per dataset would constitute test-time leakage, weakening the claim that SEER generalises across query distributions.

3.5.1 Retriever

The retriever is BM25-Okapi [1] (via the `rank_bm25` library) over the per-question candidate paragraph pool $\mathcal{P}_q$, matching the evaluation protocol of Adaptive-RAG [4] and DeepRAG [10]. Each candidate paragraph is tokenised by lower-cased word boundaries; the paragraph title is prepended to its body before indexing (titles carry strong entity signal for KG-triple queries). The retrieved context is rendered as a numbered block:

```
[1] <title-1>
<body-1>

[2] <title-2>
<body-2>
...
```

Top-k is fixed to $k = 5$ in all experiments.

3.5.2 Optional Skip-Decompose Probe

For predominantly single-hop datasets (e.g. PopQA), the decomposer can over-decompose — emitting two-step plans for one-step questions. SEER supports an optional *skip-decompose probe*: before decomposition, the answerer is asked the *original* question with retrieved context, and if its confidence signal indicates high certainty ($\sigma < \tau_{\text{skip}}$), the answer is returned directly. This branch is ablated in Chapter 4 but disabled in the main results to keep the gating policy unified.

3.6 Stage 4: Synthesis

The final answer is produced by a synthesis pass that conditions on the original question and all (s_i, a_i) pairs. The synthesis prompt encodes three answer-shaping rules learned from error analysis on multi-hop benchmarks:

1. **Comparison questions** ("which X is older / taller / earlier") return the matching *entity*, not the value being compared.

2. **Yes/No questions** ("do X and Y share a country?") return "Yes" or "No" based on agreement between sub-answers.

3. **Terse-span rule.** Strip qualifiers, dates strip to year-only when the question asks "when". This rule directly addresses the EM/F1 gap observed in pilot experiments, where verbose answers (e.g. "Conservative politician and soldier" for gold "politician") lost EM despite high F1.

The full synthesis instruction (excerpted from `seer/pipeline.py`) includes four worked few-shot examples covering comparison, Yes/No, single-fact, and "when" questions. The synthesizer emits the final span directly without any explanation, allowing standard SQuAD EM / F1 metrics to be applied without post-processing.

3.7 Training Protocol

Only the decomposer requires training; the answerer reuses the same adapter (Mode C) or runs on the base model (Mode B, see Section 3.8).

3.7.1 Backbone and LoRA Configuration

- **Backbone:** Llama-3-8B-Instruct.
- **Adapter type:** LoRA [30] on all linear modules (`lora_target=all`).
- **Rank / Alpha / Dropout:** $r = 16$, $\alpha = 32$, dropout $= 0.05$.
- **Precision:** bf16.
- **Context length:** 1024 tokens (decomposition outputs are short).

3.7.2 Training Data

Decomposer training data is constructed in the Alpaca instruction-tuning format. Three sources are used:

Table 3.2: Composition of the 12K decomposer training corpus.

Source	#samples	Decomposition source
MuSiQue	4,000	Gold sub-questions (provided by dataset)
2WikiMultihopQA	4,000	GPT-OSS-20B distillation with hop-count check
PopQA	4,000	1-step triple (`#1: subj >> prop`)
Total	12,000	

The PopQA contribution serves a specific purpose: by including single-hop samples in training, the decomposer learns that a one-step plan is a valid output, directly mitigating the over-decomposition failure mode discussed in Section 3.5. This design choice is a deliberate trade-off rather than a free improvement. Including PopQA recovers single-

hop accuracy substantially (PopQA exact-match rises from 14.2 to 39.8, Chapter 4), but it mildly degrades the multi-hop benchmarks relative to a decomposer trained on multi-hop data alone (most visibly on 2WikiMultihopQA). SEER adopts the three-source decomposer as its default precisely because it yields a *single, unified* model that performs acceptably across both single- and multi-hop query distributions without any per-dataset switching — which is the central robustness claim of this work. The accuracy cost of this unification is quantified and discussed in Chapter 4 and Chapter 5.

3.7.3 Optimisation

LoRA is trained with the LLaMA-Factory framework [31]:

- **Effective batch size:** 16 (4×4 gradient accumulation).
- **Learning rate:** 1×10^{-4} with cosine schedule, 10% warmup.
- **Epochs:** 3.
- **Total steps:** $\sim$2,250.
- **Wall time:** $\sim$38 minutes on a single A6000 (48GB) GPU.

3.8 Inference Modes

Two inference modes are evaluated to disentangle the contribution of the LoRA adapter at *answer time*:

Mode B (adapter-toggle). The LoRA adapter is enabled for decomposition and synthesis but *disabled* for the answerer probe and re-answer steps. This isolates the decomposer's contribution: the answerer behaves as pure base Llama3-8B-Instruct.

Mode C (adapter-always). The LoRA adapter is enabled for all stages. This is the default used in main results, since the decomposition LoRA does not appear to degrade answer-only generation in pilot tests.

Both modes share the same gating logic; they differ only in which weights produce

$\tilde{a}_i$ and a_i.

3.9 Implementation Summary

The end-to-end SEER inference loop is summarised as Algorithm 1. Code lives in `Research/main_methodology/seer/` (`pipeline.py`, `answerer.py`, `retriever.py`).

Algorithm 1 SEER inference for one question q

Require: question q, paragraph pool $\mathcal{P}_q$, threshold τ, top-k

$\{s_1, \ldots, s_n\} \leftarrow \mathcal{D}_\theta(q)$ ▷ decompose
$A \leftarrow \emptyset$ ▷ answers dict for placeholder resolution
for $i = 1$ to n **do**
  $s_i \leftarrow \textsc{Resolve}(s_i, A)$ ▷ substitute $\#j$ with $A[j]$
  $(\tilde{a}_i, \sigma_i) \leftarrow \mathcal{A}_\theta(s_i, \emptyset)$ ▷ probe without context
  if $\sigma_i \geq \tau$ **then**
    $c_i \leftarrow r(s_i, \mathcal{P}_q, k)$ ▷ retrieve
    $a_i \leftarrow \mathcal{A}_\theta(s_i, c_i)$ ▷ re-answer with context
  else
    $a_i \leftarrow \tilde{a}_i$
  end if
  $A[i] \leftarrow a_i$
end for
$\hat{a} \leftarrow \textsc{Synthesize}(q, \{(s_i, a_i)\}_{i=1}^{n})$
return $\hat{a}$

The next chapter evaluates this pipeline on four QA benchmarks and ablates the decomposer training data, gating signal, and threshold.

Chapter 4 Experiments

This chapter evaluates SEER on four open-domain QA benchmarks against published adaptive-RAG baselines. Section 4.1 describes datasets, metrics, and the alignment protocol with prior work. Section 4.2 reports the main results. Sections 4.3–4.5 present ablations on the gating threshold, inference mode, decomposer training data, and synthesis prompt. Section 4.7 analyses per-step retrieval trigger rates, and Section 4.8 provides per-type and per-hop breakdowns.

4.1 Experimental Setup

4.1.1 Datasets

Four QA benchmarks are evaluated, three multi-hop and one single-hop (Table 2.1 in Chapter 2). To align with the DeepRAG [10] evaluation protocol, we use the **first 1,000 questions of each dev set** (no shuffle), which exactly matches the sample slicing implemented in DeepRAG's public release. For MuSiQue, we additionally report full-dev results ($n = 2{,}417$) as a stricter evaluation.

4.1.2 Retrieval Corpus

We follow the *per-question candidate paragraph* retrieval protocol used by Adaptive-RAG [4], IRCoT [3], and the original benchmark releases: each question carries $\sim$20 candidate paragraphs (2 gold + 18 distractors for HotpotQA / 2WikiMultihopQA; MuSiQue ships its own distractor pool; PopQA passages are retrieved from a Wikipedia dump pre-filtered by entity match). BM25 ranks within this per-question pool. DeepRAG instead uses a global Wikipedia BM25 index over a DPR dump; both protocols are reported in their original papers, and the per-question setup is the more common choice in the adaptive-RAG literature.

4.1.3 Metrics

We report SQuAD-style **exact match (EM)** and token-level **F1** after standard normalisation (lowercasing, article removal, punctuation stripping). When the dataset provides multiple gold aliases (e.g. PopQA, NQ), we take the maximum over aliases. Additionally we report:

- **Retrieval rate (R%)** — the fraction of sub-question steps that trigger retrieval, measured as $\sum_q |\{i : \pi(s_i) = 1\}| \,/\, \sum_q |\mathcal{D}(q)|$.
- **Decomposition steps** — the average number of sub-questions emitted per query.

4.1.4 Baselines

We compare against six published methods plus internal control baselines:

- **No-RAG CoT.** Chain-of-thought without retrieval (parametric memory only).
- **Single-RAG.** One BM25 retrieval pass on the original query, then standard reader.
- **FLARE** [6]. Sentence-level lookahead probe; reported numbers from [10].
- **DRAGIN** [7]. Entropy + self-attention gate; reported numbers from [10].
- **UAR** [10]. Reported in DeepRAG comparison.
- **Adaptive-RAG** [4]. T5-Large query classifier.
- **DeepRAG** [10]. SFT + binary tree search + DPO (paper numbers and our 8B rerun).
- **Ours, neg_max_ent** (internal control). Pre-SEER token-entropy gate without decomposition.

4.1.5 Implementation Details

All experiments use Llama-3-8B-Instruct as the base model with a LoRA-rank-16 decomposer adapter. Unless stated otherwise, the **main results use the three-source**

(12K) decomposer trained on MuSiQue + 2WikiMultihopQA + PopQA (Chapter 3), because it yields a *single unified model* that performs acceptably across both single- and multi-hop query distributions without any per-dataset switching — the central robustness claim of this work. A two-source (8K, MuSiQue + 2WikiMultihopQA) variant, which trades single-hop coverage for stronger multi-hop accuracy, is reported as an ablation in Section 4.5. Retrieval is BM25-Okapi with $k = 5$. The unified threshold is $\tau = 0.3$ on max-entropy, and all main results use **Mode C** (LoRA adapter always on). Greedy decoding (`do_sample=False`) is used throughout for reproducibility.

4.2 Main Results

Table 4.1 compares SEER against all baselines on the aligned 1,000-question dev slice.

4.2.1 Key Findings

Finding 1: MuSiQue is a decisive win. SEER reaches 46.2 EM / 55.1 F1 on MuSiQue, exceeding the DeepRAG paper by $+23.7$ EM and $+24.7$ F1, and exceeding every other baseline by $\geq +23$ EM. This is the largest single-dataset margin among the methods compared. MuSiQue is the hardest of the four benchmarks (deeper compositional chains, harder distractors), and SEER's KG-triple decomposition appears particularly well-suited to its structured composition.

Finding 2: a single unified model is competitive everywhere, including single-hop. This is the central result. The same 12K decomposer that wins MuSiQue also reaches 39.8 EM / 48.3 F1 on the single-hop PopQA benchmark — within 2.7 EM of the specialised DeepRAG (42.5) and within 3.3 F1 of the strongest no-decomposition baseline (neg_max_ent, 51.6) — *without any per-dataset switching or routing*. No prior method in the comparison spans the single-hop / multi-hop divide with one model: DeepRAG reports PopQA only with its single-hop configuration, while our own multi-hop pipelines collapse to 14.2 EM under the two-source decomposer (Section 4.5). SEER's robustness across query distributions, at a fixed inference policy, is the contribution that the headline

Table 4.1: Main results (EM / F1 %) on the first 1,000 questions of each dev set. SEER uses the three-source (12K) decomposer with synthesis prompt v2. Best result per column in **bold**; second-best underlined.

Method	HotpotQA		2Wiki		MuSiQue		PopQA	
	EM	F1	EM	F1	EM	F1	EM	F1
Reported in DeepRAG paper [10]								
CoT (no-RAG)	27.2	37.8	28.2	34.9	13.7	23.0	–	–
FLARE [6]	23.4	32.9	30.5	37.5	17.6	23.6	–	–
DRAGIN [7]	27.6	38.1	29.1	35.7	11.8	20.0	–	–
UAR	29.7	40.7	34.8	42.4	19.1	28.4	–	–
DeepRAG	**40.7**	**51.5**	48.1	53.3	22.5	30.4	**42.5**	47.8
Our re-implementations on Llama-3-8B								
CoT (no-RAG)	25.2	35.6	31.2	39.3	11.5	20.3	20.4	23.1
Single-RAG	32.4	44.0	37.9	47.0	19.2	29.0	41.6	51.3
neg_max_ent (no decomp)	32.7	44.3	36.2	45.4	19.0	29.1	42.4	**51.6**
DeepRAG-8B + SC	39.7	50.4	**51.1**	**55.9**	22.8	30.2	–	–
SEER (Ours, 12K)	36.5	49.1	37.7	46.1	**46.2**	**55.1**	39.8	48.3

numbers should be read against.

Finding 3: the cost of unification falls on 2Wiki and HotpotQA EM. Relative to a multi-hop-specialised decomposer (8K, Section 4.5), the unified model gives up EM on 2Wiki ($45.0 \rightarrow 37.7$) and HotpotQA ($39.6 \rightarrow 36.5$), and consequently trails DeepRAG on these two. Part of this gap is span-trimming (factually correct spans carrying extraneous modifiers, addressed by synthesis prompt v2 in Section 4.6) and part is genuine under-decomposition induced by the single-hop training signal. This trade-off is quantified in Section 4.5 and discussed as the principal open problem in Chapter 5.

4.3 Ablation: Gating Threshold τ

The threshold τ controls the retrieval / parametric trade-off. Table 4.2 sweeps $\tau \in [0, \infty)$ on MuSiQue (200-question sample) with the max-entropy signal.

Table 4.2: Threshold sweep of τ on max-entropy signal (MuSiQue, $n = 200$). Retrieval% is the fraction of sub-question steps that trigger retrieval.

τ	Retrieval%	EM	F1	Notes
0.0 (always)	100.0	47.5	58.9	Upper bound
0.2	94.0	47.5	58.9	Gate matches always-RAG
0.3	90.8	47.0	58.8	**Sweet spot**
0.5	87.5	46.5	58.0	-1 EM
0.8	78.1	41.5	54.3	Cliff begins
1.0	75.1	41.0	53.6	-6 EM
1.5	65.8	36.5	51.3	-11 EM
2.5	50.1	27.0	41.6	-20 EM
3.5	26.7	18.5	30.1	-29 EM
∞ (never)	0.0	7.5	18.2	Lower bound

The curve exhibits a sharp **cliff above** $\tau = 0.5$: as long as retrieval rate stays $\geq 80\%$, EM is within 1 point of the always-retrieve upper bound, while below this rate accuracy degrades almost linearly. The cliff position is consistent with the pre-SEER `neg_max_ent@-0.80` observation that the entropy distribution of correct-answer probes concentrates near zero. This justifies the choice of $\tau = 0.3$ as a single dataset-agnostic threshold — it sits well inside the safe region while still saving 7–9 percentage points of retrieval cost (Section 4.7).

4.4 Ablation: Inference Mode B vs C

Table 4.3 compares Mode B (LoRA off during answering) and Mode C (LoRA on throughout) on MuSiQue 200.

Table 4.3: Inference mode ablation (MuSiQue, $n = 200$).

Mode	Description	EM	F1	Retrieval%
B	Decompose / synth with LoRA; answer with base model	35.0	47.6	71.1
C	LoRA enabled at all stages	**40.5**	**53.3**	69.6

Mode C wins by $+5.5$ EM and $+5.7$ F1 with effectively identical retrieval rate. The decomposition adapter does not degrade answer-only generation; on the contrary, it slightly improves it, likely because the adapter has been exposed to entity-property patterns during training that overlap with the answerer's task. **All subsequent experiments use Mode C.**

4.5 Ablation: Decomposer Training Data

SEER's default decomposer is trained on the 12K three-source mix (MuSiQue + 2Wiki + PopQA, "3DS"). To isolate the effect of the single-hop component, we ablate it: a 8K two-source decomposer (MuSiQue + 2Wiki only) is trained under identical hyperparameters. Table 4.4 reports both.

The single-hop component is what makes SEER a unified model: it lifts PopQA by $+25.6$ EM (from a broken 14.2 to a competitive 39.8) by teaching the decomposer that a *one-step plan is a valid output*, directly mitigating the over-decomposition failure mode. The cost is multi-hop accuracy: 2Wiki loses -7.3 EM, HotpotQA -3.1, and MuSiQue -1.8, because the same single-hop signal also pushes the decomposer to under-decompose genuine 2-hop questions. MuSiQue is the most robust to the change — its chains are deeper and the KG-triple supervision is internally consistent. This is a clear

Table 4.4: Effect of removing the 4K PopQA single-hop component from the decomposer training corpus. All values are EM / F1 on 1000-question dev slices. The 12K three-source mix is SEER's default.

Training mix	HotpotQA	2Wiki	MuSiQue	PopQA
12K three-source (default)	36.5 / 49.1	37.7 / 46.1	46.2 / 55.1	39.8 / 48.3
8K two-source (no PopQA)	39.6 / 52.4	45.0 / 53.7	48.0 / 57.1	14.2 / 21.8
Δ (12K $-$ 8K)	-3.1 / -3.3	-7.3 / -7.6	-1.8 / -2.0	**+25.6 / +26.5**

example of negative transfer: **single-hop and multi-hop decomposition supervision interfere.** SEER adopts the 12K mix as default because the robustness of a single model across the full query distribution outweighs the multi-hop EM it costs; closing that gap — e.g. by conditioning the decomposer on a length-target token, analogous to length control in summarisation, so one model can emit different decomposition depths on demand — is the principal future direction (Chapter 5).

4.6 Ablation: Synthesis Prompt

The terse-answer rules and worked few-shot examples in the synthesis prompt (Section 3.6) were added in response to the observed EM/F1 gap on 2Wiki bridge-comparison questions. The change is *prompt-only* — no re-training — so its effect is independent of the decomposer mix; the measurements below were taken on the two-source decomposer during development and transfer to the default model. PopQA is omitted: it has a single sub-step, so there is nothing to synthesise and the prompt has no effect.

The 2Wiki gain ($+9.1$ EM) is the headline: it derives almost entirely from the bridge-comparison subtype, where the original synthesizer emitted KG-triple-shaped strings (e.g. `X >> date of birth`) instead of resolving the comparison. The few-shot examples teach the model to map "compare two dates" to "return the entity owning the later date." MuSiQue and HotpotQA also benefit moderately. **This is a free win: no re-training, no regression, every multi-hop benchmark improved.**

Table 4.5: Effect of upgrading the synthesis prompt from v1 (rules only) to v2 (rules + comparison + Yes/No few-shot), measured on the multi-hop benchmarks. EM % on 1000-question slices.

Prompt	HotpotQA	2Wiki	MuSiQue
v1 (rules only)	37.8	35.9	45.9
v2 (rules + few-shot)	39.6	45.0	48.0
Δ	$+1.8$	**+9.1**	$+2.1$

4.7 Retrieval Trigger Rate Analysis

Table 4.6 reports per-dataset retrieval trigger rates under the default $\tau = 0.3$.

Table 4.6: Retrieval rate at $\tau = 0.3$ (max-entropy gate). $|\mathcal{D}|$ is average decomposition length.

Dataset	$\lvert\mathcal{D}\rvert$ (avg steps)	Retrieve / Total steps	Trigger Rate
HotpotQA	2.26	2060 / 2257	91.3%
2Wiki	2.47	2357 / 2472	95.3%
MuSiQue	2.03	1832 / 2032	90.2%
PopQA	1.00	993 / 1000	99.3%

Three observations:

1. **Trigger rates cluster between 89–95% for multi-hop benchmarks.** The gate does not skip aggressively, which is consistent with the threshold-sweep cliff at 80% (Section 4.3): SEER deliberately operates in the safe region above the cliff.

2. **2Wiki has the highest trigger rate (95.3%)**, reflecting that its Wikidata-derived questions reference long-tail entities the base Llama-3 has weak parametric coverage of.

3. **PopQA triggers retrieval on nearly every step (99.3%)** because its questions hinge on long-tail entity facts where parametric memory is unreliable. The high

trigger rate is not the cause of PopQA's low EM — the cause is over-decomposition before retrieval is even invoked.

The gate therefore saves 5–11 percentage points of retrieval calls compared to always-retrieve, while losing ≤ 0.5 EM (Table 4.2) — a clear Pareto improvement.

4.8 Per-Type and Per-Hop Breakdowns

4.8.1 2Wiki by Question Type

Table 4.7 breaks down 2Wiki performance by official question type, exposing where the bulk of the synthesis-prompt gain came from.

Table 4.7: 2Wiki per-type EM (1000-question slice, $\tau = 0.3$, synthesis prompt v1 vs v2).

Type	n	v1 EM	v2 EM	Δ
inference	128	58.6	60.2	$+1.6$
compositional	428	46.3	50.0	$+3.7$
comparison	216	29.6	41.7	$+12.1$
bridge_comparison	228	9.6	39.5	**+29.9**
Total	1000	35.9	45.0	$+9.1$

Bridge-comparison (4-step double-chain questions) went from **9.6% to 39.5% EM**, a near $4\times$ improvement attributable entirely to the synthesis prompt v2. Inference and compositional questions, which were already working, see modest gains.

4.8.2 MuSiQue by Hop Depth

MuSiQue's full dev set (2,417 questions) is partitioned by hop depth in Table 4.8.

Performance is monotonically decreasing in hop count, as expected. The 4hop variants are the principal short-coming and align with the decomposer count-match ablation

Table 4.8: MuSiQue full-dev EM by hop class (2,417 questions, $\tau = 0.3$).

Hop class	n	EM
2hop	1252	**43.0**
3hop_v1	568	36.6
3hop_v2	192	15.1
4hop_v1	246	25.6
4hop_v2	64	14.1
4hop_v3	95	9.5
Total	2417	35.4

in Chapter 3: the decomposer's count-match drops to 40–46% on 4-hop chains, propagating errors into both retrieval and synthesis. The first 1,000 questions of MuSiQue dev consist entirely of 2-hop examples (the dataset ships sorted by hop count), which is why the aligned-protocol number (48.0 EM) is higher than the full-dev number (35.4 EM); we report both in the interest of fairness.

4.9 Summary

The experiments support three conclusions:

- **One unified model spans single- and multi-hop.** The default three-source (12K) SEER decisively wins MuSiQue ($+23.7$ EM over DeepRAG) while remaining competitive on single-hop PopQA (39.8 EM, within 2.7 of specialised DeepRAG; 48.3 F1, above it) — with no per-dataset routing. This robustness across the query distribution is the central contribution.

- **The gating signal is robust.** A single threshold $\tau = 0.3$ on max-entropy works across all four datasets without per-dataset tuning, and the threshold-sweep cliff at 80% retrieval rate justifies the choice.

- **Unification has a measurable cost on 2Wiki / HotpotQA EM.** Against a multi-

hop-only (8K) decomposer, adding single-hop supervision rescues PopQA by $+25.6$ EM but costs 7.3 EM on 2Wiki and 3.1 on HotpotQA — a clear sign of supervision interference. Closing this gap with a single model is the principal open problem, discussed at length in Chapter 5.

Chapter 5 Discussion

This chapter interprets the experimental findings of Chapter 4. Section 5.1 compares SEER and DeepRAG along the training-cost / accuracy frontier. Section 5.2 dissects the EM–F1 gap and characterises the residual failure modes. Section 5.3 analyses the over-decomposition limitation exposed by PopQA. Section 5.4 discusses what the threshold sweep tells us about the entropy signal itself. Section 5.5 acknowledges threats to validity.

5.1 Cost–Accuracy Trade-off versus DeepRAG

DeepRAG [10] and SEER occupy adjacent points on the cost–accuracy frontier of adaptive RAG. The pipelines differ in three quantitative dimensions:

Table 5.1: Approximate training cost comparison between SEER and DeepRAG, normalised to a single A6000 (48GB) GPU.

Component	SEER	DeepRAG
Training data	8K–12K Alpaca samples	5K seeds $\rightarrow\approx$10K via tree search
Training stages	1 (LoRA SFT)	2 (SFT + DPO / Chain of Calibration)
Tree-search rollouts	–	$O(2^H)$ branches per question
Preference pair construction	–	required for DPO
Wall-clock (LoRA)	$\sim$38 min	multi-hour SFT + multi-hour DPO
Gate signal	token entropy (training-free)	learned implicit policy
Interpretability	scalar σ + threshold τ	opaque

The key trade-offs that fall out of these differences are:

1. **SEER wins on training simplicity.** A single LoRA SFT pass on 8K samples is approximately one order of magnitude cheaper than DeepRAG's full pipeline, and avoids the engineering risk of constructing well-balanced DPO preference pairs.

2. **DeepRAG wins on single-hop (PopQA).** Its Stage-II Chain of Calibration explicitly teaches the model when *not* to retrieve and when to emit short answers — both of which directly help PopQA's structure. SEER's single SFT pass cannot mimic this without supervision interference (Section 5.3).

3. **SEER ties or wins on multi-hop F1.** On HotpotQA, 2Wiki, and MuSiQue, SEER's F1 matches or exceeds DeepRAG despite a much cheaper pipeline. This suggests that when the dominant bottleneck is *reasoning composition*, the gating signal matters less than the decomposition quality.

4. **SEER wins on interpretability.** The entropy threshold τ is a single scalar a practitioner can inspect, tune, and monitor in production; DeepRAG's gate is a learned implicit policy with no externally legible score.

The headline trade-off is therefore: *SEER is cheaper, more interpretable, and competitive on multi-hop reasoning; DeepRAG is more accurate on single-hop fact lookup at substantially higher training cost.*

5.2 The EM–F1 Gap and Residual Failure Modes

A recurring observation in Chapter 4 is that SEER's F1 often exceeds the corresponding baseline while its EM lags. The synthesis prompt v2 closed most of this gap on 2Wiki (Table 4.5), but a residual gap of 1–3 EM persists on HotpotQA and 2Wiki. Manual inspection of $\sim$60 failure cases identifies three patterns:

5.2.1 Verbose Spans with Correct Information

The most common pattern (estimated 40–50% of EM-only failures): the predicted span contains the gold answer but adds an apposition or qualifying clause. Examples:

These failures could be addressed by a stricter post-processor (e.g. first-NP extraction), but doing so risks dropping the modifier when it *is* the correct answer ("Conservative Party" vs "politician"). The synthesis prompt v2 partially addresses the issue via

Table 5.2: Representative EM-only failures where the predicted span subsumes the gold but is too long.

Question	Gold	SEER prediction
What is George Rankin's occupation?	politician	Conservative politician and soldier
Where did the director of *X* grow up?	London	London, England
What disease did the patient have?	tuberculosis	extrapulmonary tuberculosis

the "KEEP IT TERSE" rule and "strip modifiers" few-shot example; further gains likely require either RL with EM-shaped reward or supervised post-training on terse spans (i.e. approximately what DeepRAG's Chain of Calibration does).

5.2.2 Alias Mismatch

PopQA contributes many failures of this kind: the model emits "Bill Clinton" for gold "William Jefferson Clinton", or "USA" for gold "United States of America". The dataset provides alias lists but they are not exhaustive. An LLM-as-Judge evaluator (e.g. GPT-4 acting as semantic-equivalence judge) would likely raise reported EM by several points without any change to the model itself; we do not include this in the main results to avoid the additional methodological debate.

5.2.3 Wrong Sub-Answer Propagation

A smaller but more concerning failure: an incorrect sub-answer to step i propagates through $\#i$ to subsequent steps. Once the bridge entity is wrong, the rest of the chain is wrong regardless of retrieval quality. This is intrinsic to a sequential decomposer; mitigations include (i) re-ranking top-k candidates by downstream answer plausibility, (ii) emitting multiple candidate decompositions and selecting the highest-confidence final answer, or (iii) self-consistency aggregation across decoding samples. None of these are implemented in the present SEER and they constitute the most promising direction for raising multi-hop EM.

5.3 The Over-Decomposition Limitation (PopQA)

The PopQA result is the largest single weakness of SEER and deserves dedicated analysis. Three observations follow from the training-data ablation (Table 4.4):

1. **The decomposer is the bottleneck, not retrieval.** PopQA's retrieval trigger rate is 99.3% — the gate fires on essentially every step. Yet 8K-trained EM is only 17.9. The accuracy ceiling is therefore set by what happens *before* retrieval: the decomposer's choice of step count and triple structure.

2. **Single-hop and multi-hop supervision interfere.** Mixing 4K PopQA single-hop samples lifts PopQA EM from 17.9 to 37.3 ($+19.4$) but costs 2Wiki -5.7 and HotpotQA -3.5. The model cannot serve both regimes from a single LoRA without conditional guidance.

3. **Entropy-based skip-decompose is not a unified solution.** A preliminary experiment with a "skip if the original question can be answered confidently with retrieval" bypass (probe with $\tau_{\text{skip}} = 2.0$ on max-entropy before decomposing) rescued PopQA by $+12.4$ EM but *simultaneously* dropped MuSiQue by -17.9 and 2Wiki by -15.6. The model is false-confident on many multi-hop questions because the first sub-question is often easy, and a single-point probe cannot distinguish "easy first hop of a hard question" from "easy single-hop question".

The right fix appears to be *conditional decomposition*: explicitly tell the decomposer how many steps to emit, either via a control token (`[1step]`, `[2step]`, ...) trained jointly with both distributions, or via a small dataset-classifier upstream of the decomposer (similar in spirit to Adaptive-RAG's complexity classifier but with finer granularity). Both are deferred to future work.

5.4 What the Threshold Sweep Reveals About the Signal

The threshold sweep (Table 4.2) is more than an ablation — it diagnoses the geometry of the entropy distribution itself. Three features stand out:

- **A wide plateau between $\tau = 0$ and $\tau = 0.5$** (-1 EM, -13 percentage points of retrieval). The model is confident enough on the easy steps that we can safely skip retrieval there without harming accuracy. This confirms the basic premise of entropy-gated RAG: there *is* a distinguishable population of confident steps.

- **A cliff between $\tau = 0.5$ and $\tau = 1.0$** (-6 EM, -12 percentage points of retrieval). Once τ rises into the bulk of the entropy distribution, every additional skipped retrieval is increasingly costly. This indicates that the confident and uncertain populations are not cleanly separable — they overlap in the $[0.5, 1.0]$ range.

- **A linear decay above $\tau = 1.0$** approaching the no-RAG lower bound. In this regime the gate is essentially flipping coins, and EM falls roughly in proportion to retrieval coverage.

The plateau–cliff–decay shape suggests that max-entropy alone is necessary but not sufficient: it captures the easy tail well but cannot disambiguate the difficult middle. A natural extension is to combine max-entropy with a second orthogonal signal (e.g. self-consistency variance across multiple decoding samples, or top-1/top-2 logit margin) to push the cliff further right. This would be the most direct path to lowering retrieval cost without losing EM, and is the strongest future-work proposal arising from the present experiments.

5.5 Threats to Validity

We discuss four potential threats:

5.5.1 Retrieval Corpus Mismatch

SEER and DeepRAG use different retrieval corpora: SEER ranks per-question candidate paragraphs (Adaptive-RAG / IRCoT convention), while DeepRAG ranks the global DPR Wikipedia dump. The per-question setup has a higher ceiling because the gold paragraph is guaranteed to be in the pool, which slightly favours SEER. We mitigate this by reporting our own re-implementations of CoT, Single-RAG, and `neg_max_ent` under the

same per-question protocol — so the relative ordering of methods within the SEER block of Table 4.1 is on equal footing. Direct head-to-head with DeepRAG should be read with this caveat.

5.5.2 Dataset Slicing

The first-1000 protocol exposes a known artefact of MuSiQue dev (sorted by hop, so the first 1000 are all 2-hop). We address this by reporting full-dev results (35.4 EM on 2,417 questions) alongside the aligned-protocol 48.0 EM. DeepRAG's own paper reports against the same slicing; head-to-head numbers are therefore consistent, but *absolute* multi-hop difficulty is understated in the aligned protocol.

5.5.3 Single Seed / Greedy Decoding

All reported numbers use greedy decoding (`do_sample=False`) with no seed averaging. Variance estimates would strengthen claims of small EM differences ($\pm$1–2 EM). However the principal SEER claims (MuSiQue $+25.5$ EM, F1 leadership on three datasets, the threshold-sweep cliff shape) are well outside any plausible single-seed noise band.

5.5.4 Decomposer Distribution Shift

The decomposer is trained on a mixture of three datasets but evaluated on a fourth (HotpotQA) it has not seen at training time. The HotpotQA results therefore demonstrate out-of-distribution generalisation of the decomposition adapter. While the 12K-mix experiments show that adding PopQA training data hurts HotpotQA, the 8K-mix (no HotpotQA training) achieves competitive HotpotQA performance (39.6 EM, only -1.1 vs DeepRAG), which we read as positive evidence for the cross-dataset transferability of KG-triple decomposition.

Chapter 6 Conclusions and Future Work

6.1 Summary of Contributions

This thesis introduced **SEER**, a sub-question-level adaptive retrieval-augmented generation framework that combines a lightweight LoRA decomposer with a training-free entropy gate. The principal contributions are:

1. **KG-triple decomposition for adaptive RAG.** We propose representing each sub-question as an `entity >> relation` fragment with `#N` placeholders for cross-step dependencies. This format is compact, consistent across datasets, and semantically aligned with the entropy probe that follows — the probed answer span is exactly the object entity of the triple.

2. **Per-sub-question entropy gate with a unified threshold.** We show that maximum per-token entropy over the tentative answer span is a reliable confidence signal, and that a single dataset-agnostic threshold $\tau = 0.3$ saves 5–11% of retrieval calls without sacrificing accuracy. The threshold sweep characterises the plateau–cliff geometry of the entropy distribution and identifies the safe operating region.

3. **A single unified model competitive at an order-of-magnitude lower training cost.** Using only a short LoRA SFT pass on the three-source (12K) corpus plus a hand-designed synthesis prompt, one SEER model decisively wins MuSiQue ($+23.7$ EM over published DeepRAG) while remaining competitive on single-hop PopQA (39.8 EM, within 2.7 of specialised DeepRAG; 48.3 F1, above it) — with no per-dataset routing. No DPO, no tree search, no preference pairs are required.

4. **An interpretable gating signal.** Unlike learned policies, the entropy threshold is a single scalar a practitioner can inspect, tune per deployment, and monitor in production. The threshold-sweep curve also serves as a built-in calibration tool for new datasets.

5. **A diagnosed structural limitation.** We identify and characterise the over-decomposition failure mode on single-hop benchmarks, show via ablation that naive single-hop / multi-hop supervision mixing causes negative transfer, and articulate concrete paths to a unified solution.

6.2 Limitations

The principal limitations of the present SEER are:

- **Residual single-hop gap and its multi-hop cost.** On PopQA the unified (12K) model reaches 39.8 EM, trailing specialised DeepRAG by only 2.7 EM (and slightly leading it in F1, 48.3 vs 47.8), but still short of a plain single-RAG baseline. Including the single-hop training signal that closes most of this gap costs 3–7 EM on 2Wiki / HotpotQA — the negative-transfer trade-off (Section 5.3). The entropy gate is not at fault; the open problem is achieving both regimes in one model without interference.

- **Deep-chain degradation.** On 4-hop MuSiQue subsets, EM drops to 9.5–14.1%. The decomposer's count-match drops below 50% at four hops, and errors propagate through the chain.

- **Sequential dependency on first hops.** Wrong sub-answers at early steps poison everything downstream via `#N` substitution. SEER has no re-ranking or self-consistency mechanism today.

- **Entropy alone is insufficient at the difficult middle.** The threshold sweep shows that entropy cleanly separates the easy tail but overlaps on harder steps. A second orthogonal signal is needed to push the gate's Pareto frontier further.

- **Greedy decoding only.** All reported numbers are single-seed greedy decoding. Variance bands would strengthen small EM differences but are not reported.

6.3 Future Work

Five directions follow directly from the limitations and discussion:

6.3.1 Length-Conditioned Decomposer

The cleanest remedy to over-decomposition is to make the decomposer conditional on a target depth. Concretely: prepend a control token `[K=1]`, `[K=2]`, ..., `[K=4]` during training, and supply the token at inference based on (i) an upstream complexity classifier (Adaptive-RAG-style) or (ii) iterative deepening with a confidence stopping rule. This would let a single LoRA serve both single-hop and multi-hop regimes without the negative transfer observed in Table 4.4.

6.3.2 Hybrid Confidence Signals

The plateau–cliff–decay shape of the threshold curve (Section 5.4) indicates that max-entropy alone cannot disambiguate the difficult middle. Three complementary signals are worth combining:

- **Self-consistency variance.** Sample N generations and measure inter-sample disagreement; high variance implies low confidence regardless of per-step entropy.
- **Top-1 / Top-2 logit margin.** Captures cases where the model is committed to one wrong token.
- **Hidden-state variance.** As proposed by SeaKR [8], the residual stream often carries factual uncertainty not visible at the output layer.

A learned linear combination of these signals (with $\sim$100 calibration labels per dataset) is a low-cost intervention that should push the cliff further right.

6.3.3 Multi-Sample Decomposition with Re-Selection

Sampling $N = 5$ decompositions and selecting the one with the highest end-to-end synthesised-answer confidence (or self-consistency across samples) is a natural comple-

ment to entropy gating. It directly addresses the sub-answer propagation failure mode (Section 5.2) by allowing the pipeline to recover from a wrong first hop.

6.3.4 Span-Trimming Calibration

The residual 1–3 EM gap on multi-hop datasets is largely verbosity. A small EM-shaped reward fine-tune (or alternatively a Chain-of-Calibration-style DPO pass on terse vs verbose pairs) would close this gap. This would amount to importing DeepRAG's Stage II selectively, applied only to the synthesis pass — much cheaper than full DPO.

6.3.5 Beyond English Wikipedia

All experiments use English Wikipedia-derived benchmarks. Extending SEER to multilingual or domain-specific corpora (legal, biomedical, code) would test the cross-domain generality of KG-triple decomposition and entropy gating. The training-free gate is particularly attractive in low-resource settings where DPO preference data is unavailable.

6.4 Closing Remarks

Adaptive retrieval-augmented generation has rapidly converged toward sub-question-level decision making. The two leading designs — SEER (this work) and DeepRAG — represent opposite ends of the training-cost spectrum, and the present results suggest that the cheaper, more interpretable end can serve both single- and multi-hop queries with a single unified model. The remaining EM gap on 2Wiki / HotpotQA is real but structurally addressable; the future-work proposals above describe concrete paths forward.

More broadly, we view this thesis as evidence that a well-chosen *representational format* (the KG-triple decomposition) combined with an *interpretable scalar gate* (max-entropy with threshold τ) can substitute for substantial training machinery. As retrieval-augmented systems become embedded in production assistants, that combination of low cost, transparency, and competitive accuracy is likely to be increasingly valuable.

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Appendices

Appendix A Supplementary Material

This appendix contains supplementary material such as full prompt templates, additional experimental results, dataset statistics, hyperparameter settings, and source code snippets.